Carnegie Mellon University

GRT

Towards Foundational Models for Single-Chip Radar

ICCV 2025

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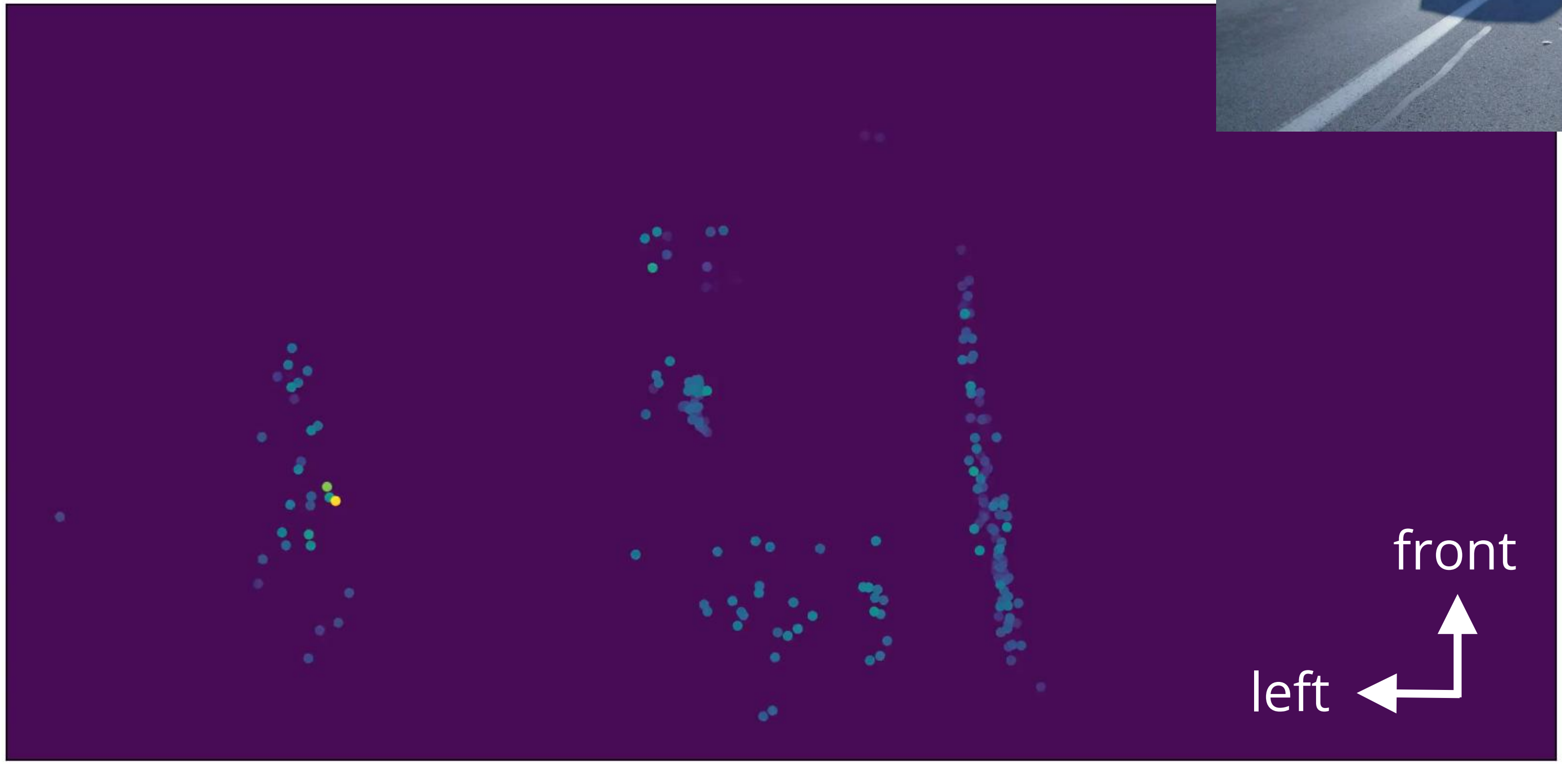
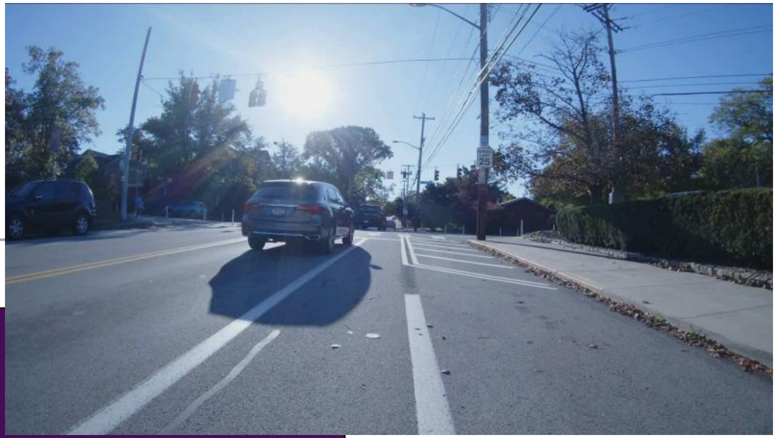
In this talk: **we turn this...**

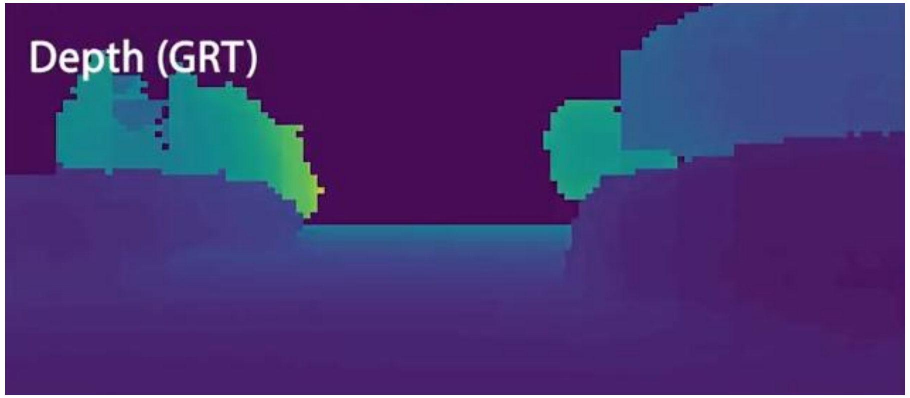


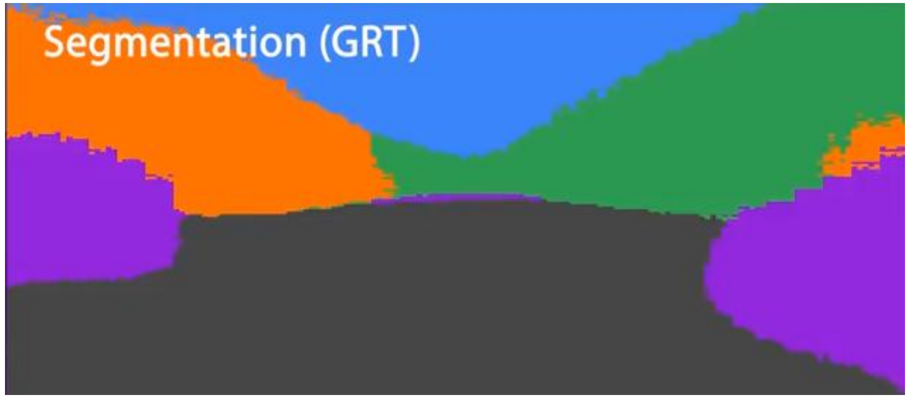


camera reference

… into this

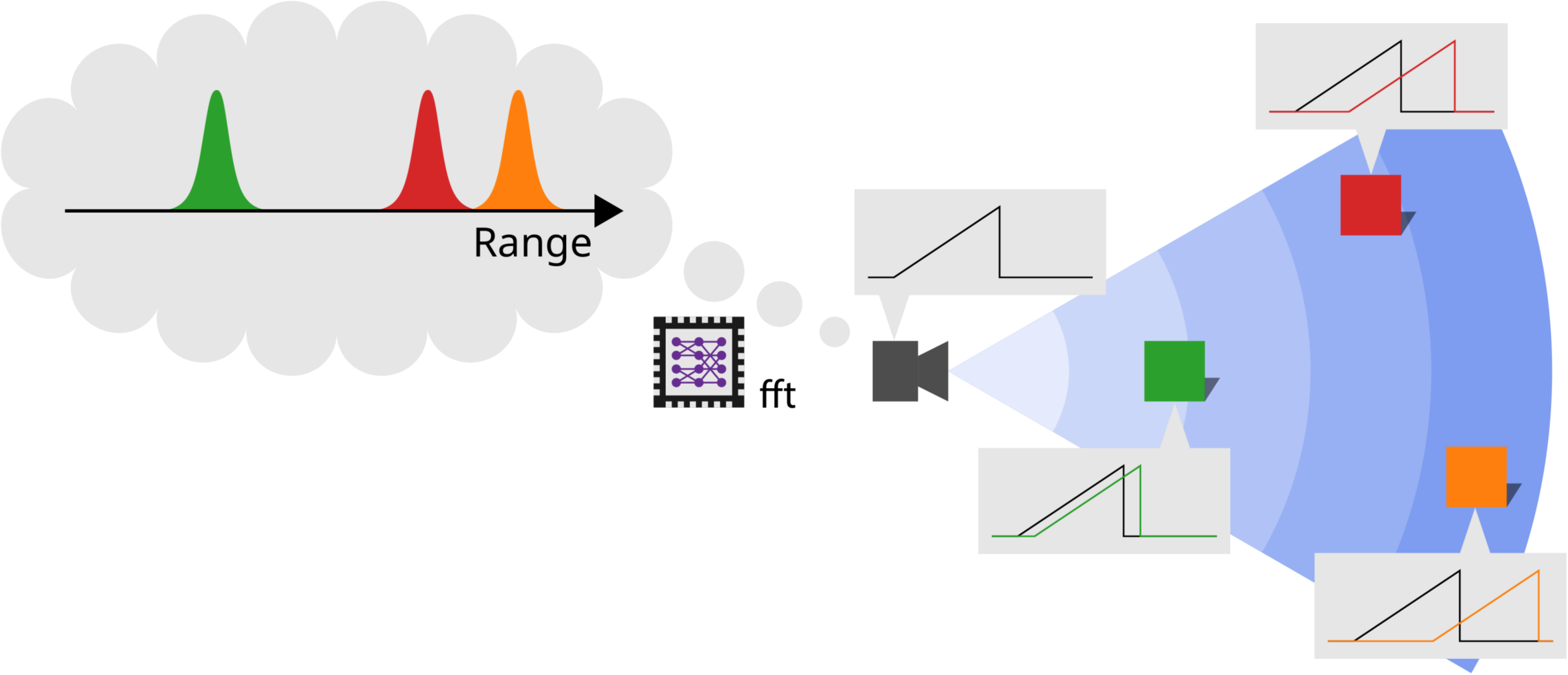

RGB (Ground Truth)


Depth (GRT)


Segmentation (GRT)


Birds Eye View Occupancy (GRT)

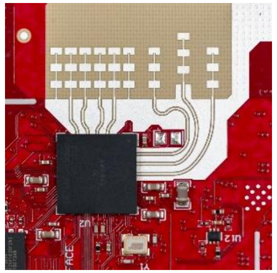
Radar: Time of Flight Ranging



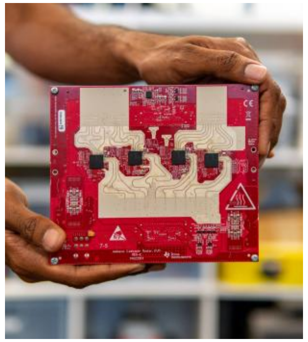
Why Radar?
Radar vs Cameras

Why Radar?
Radar vs Lidar

$** **$$** **$$$+


Single-Chip Radar

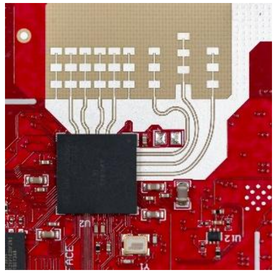
Cameras


Imaging Radar


FMCW Lidar

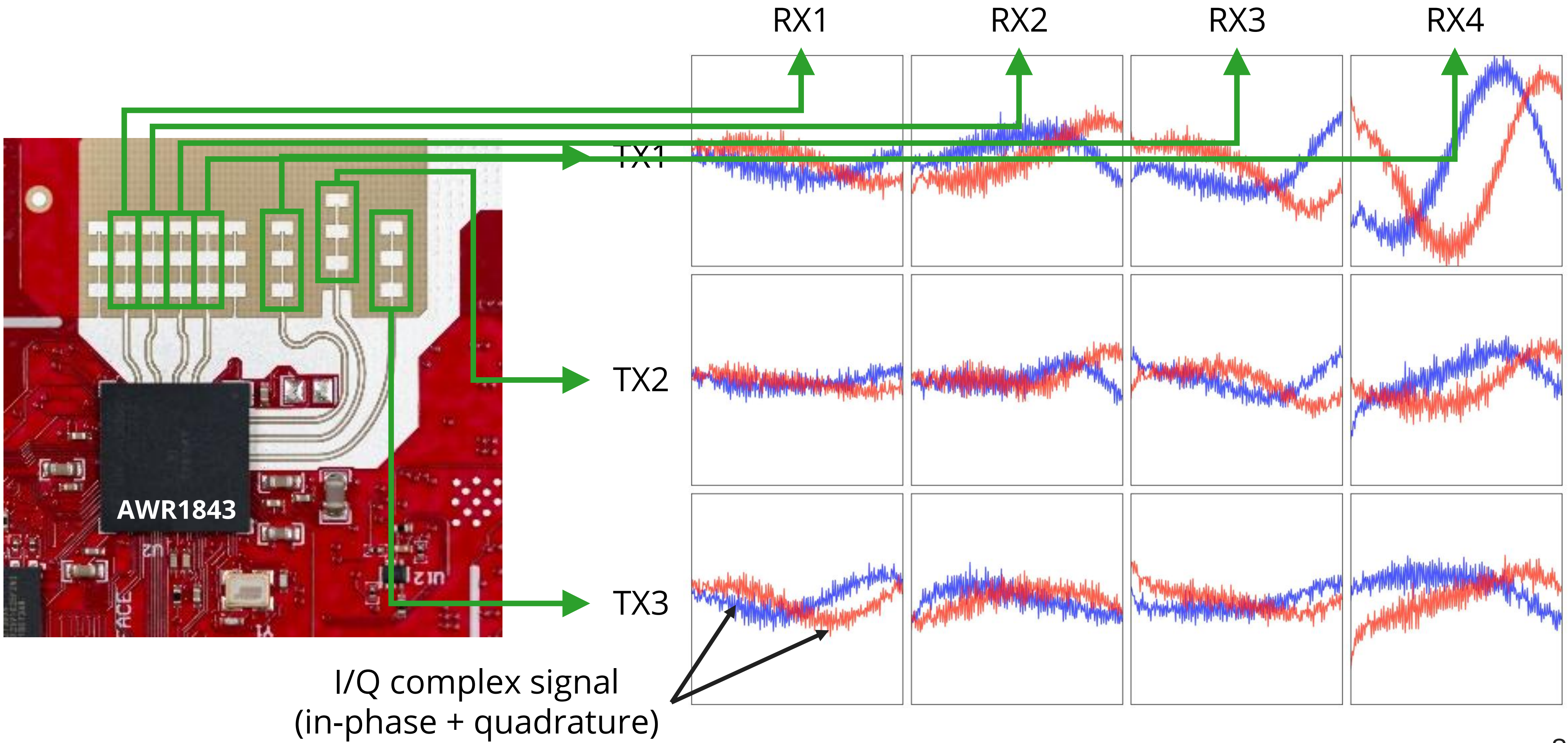

Traditional Lidar

Why Radar?

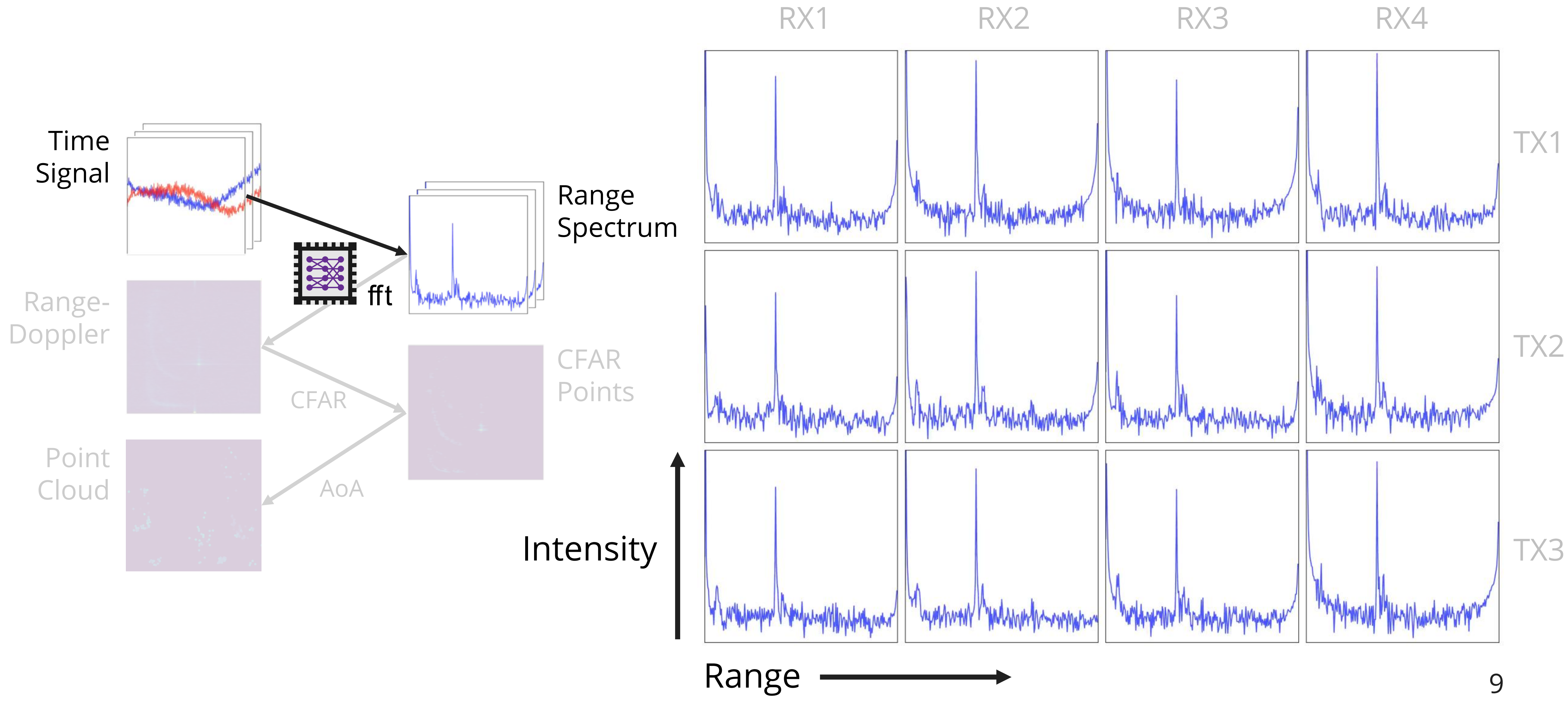


	Radar	Camera	Lidar
Angular Ambiguity	✗ High	✓ Low	✓ Low
Spatial Resolution	✗ Low	✓ High	✓ High
Range Ambiguity	✓ Low	✗ High	✓ Low
Works in the Dark	✓ Yes	✗ No	✓ Yes
Robustness to Occlusion	✓ High	✗ Low	✗ Moderate
Cost	$	$	$$$

Radar Processing: **Time Signal**



Radar Processing: **Range Spectrum**



Radar Processing: **Range-Doppler Spectrum**

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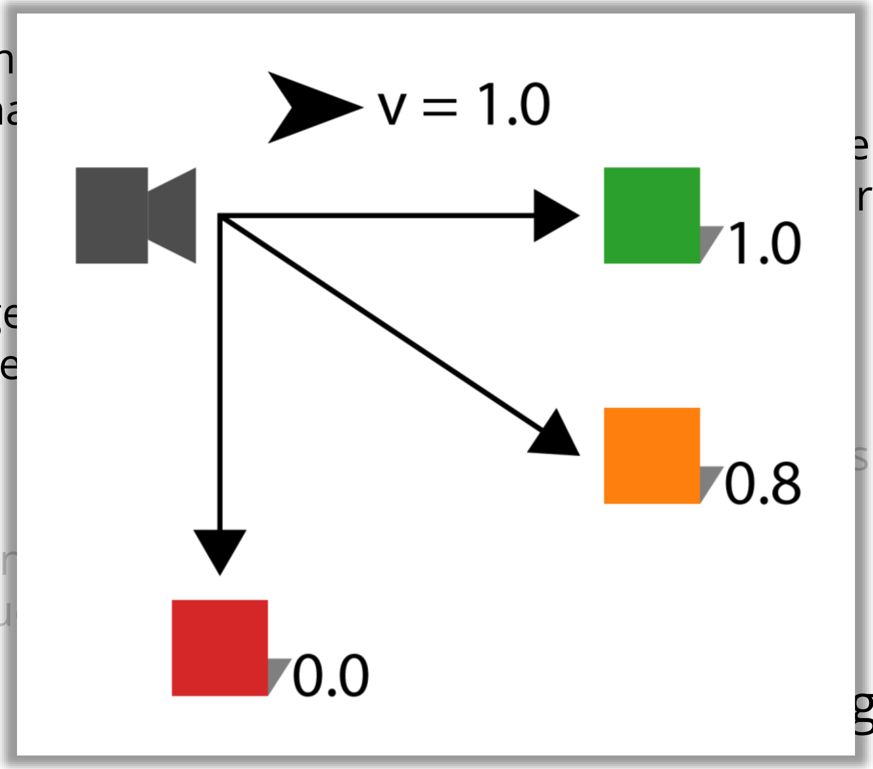
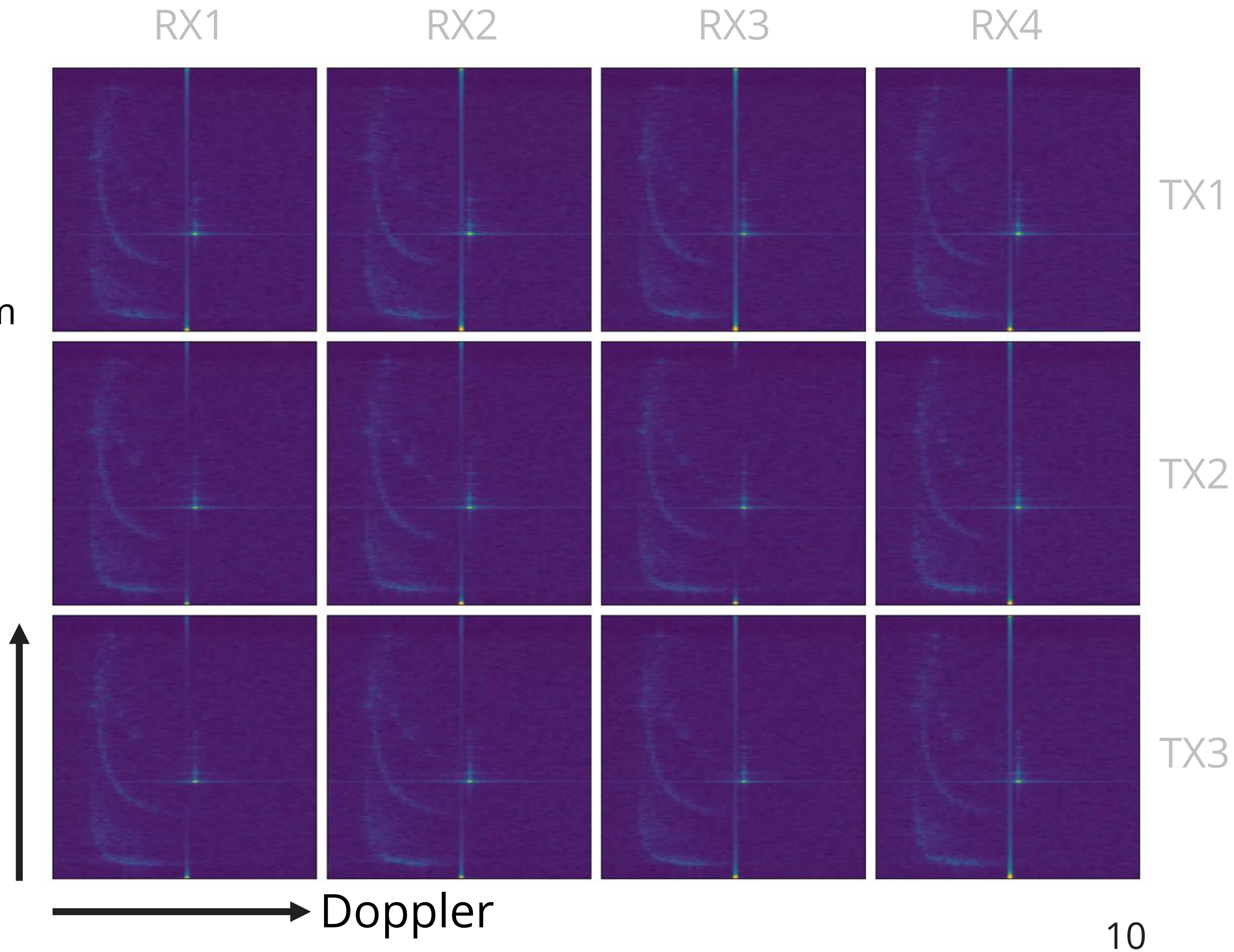
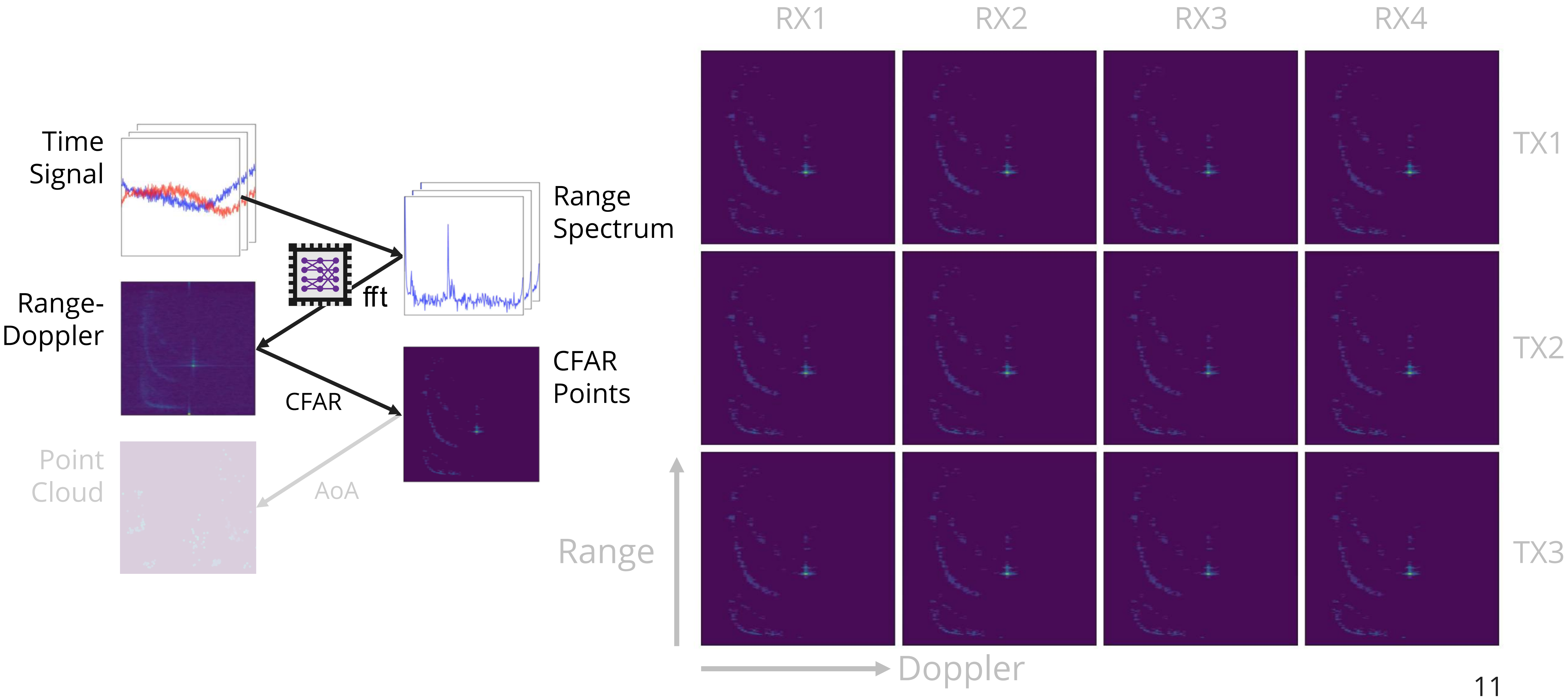
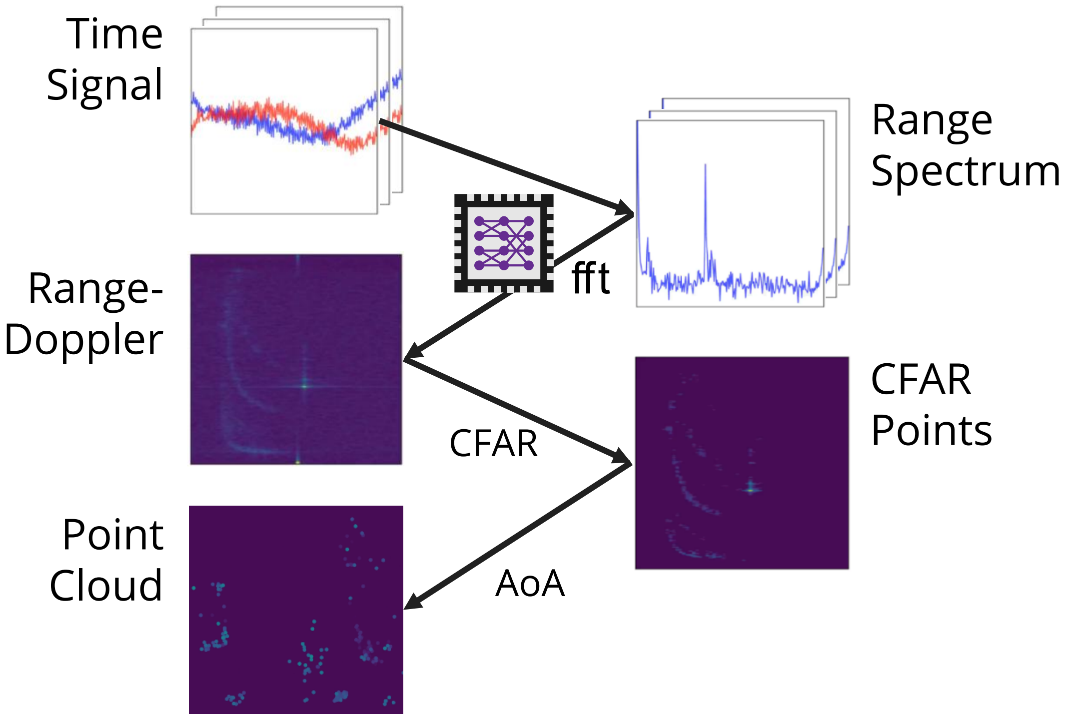


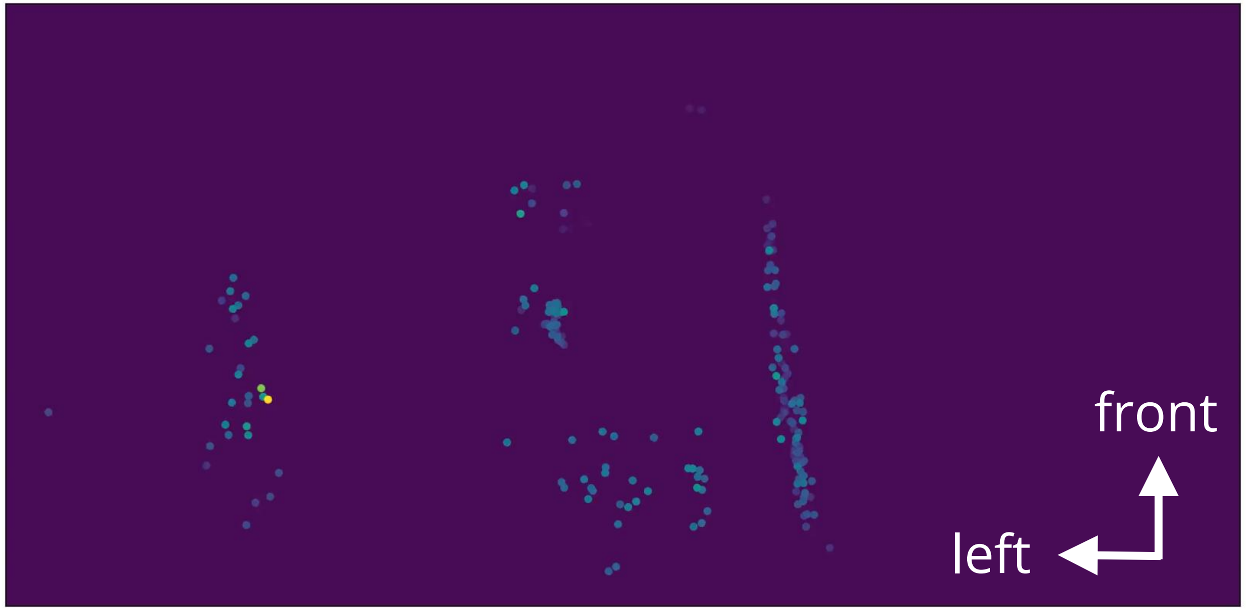


Radar Processing: **CFAR Peak Detection**

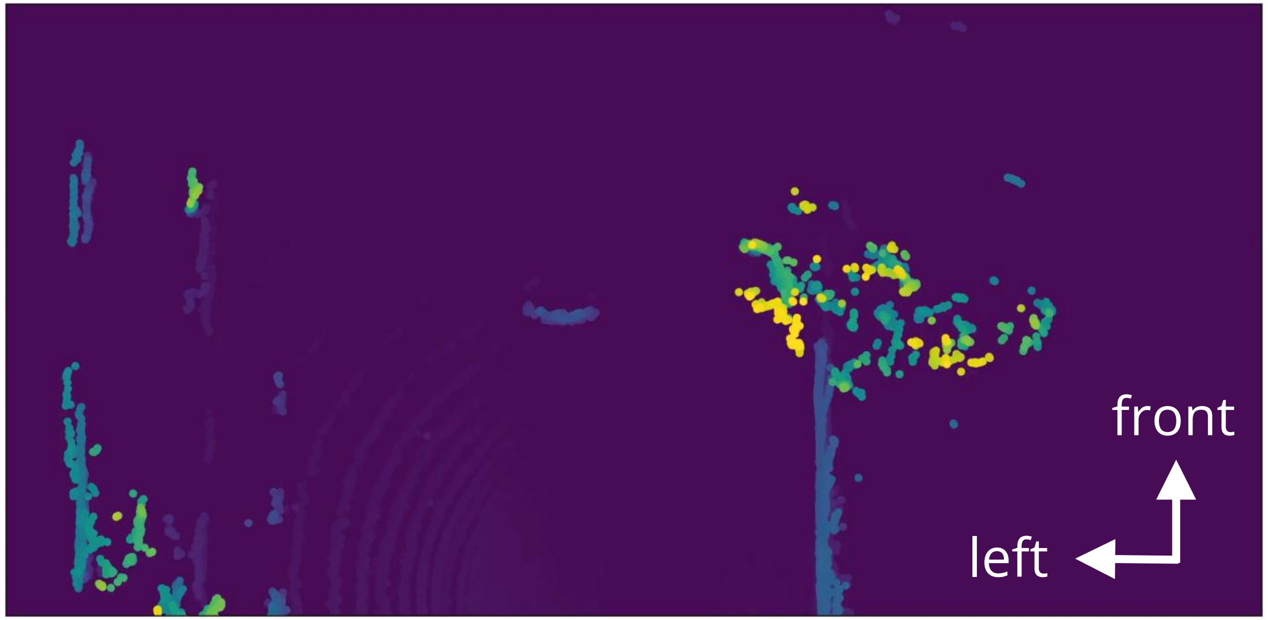


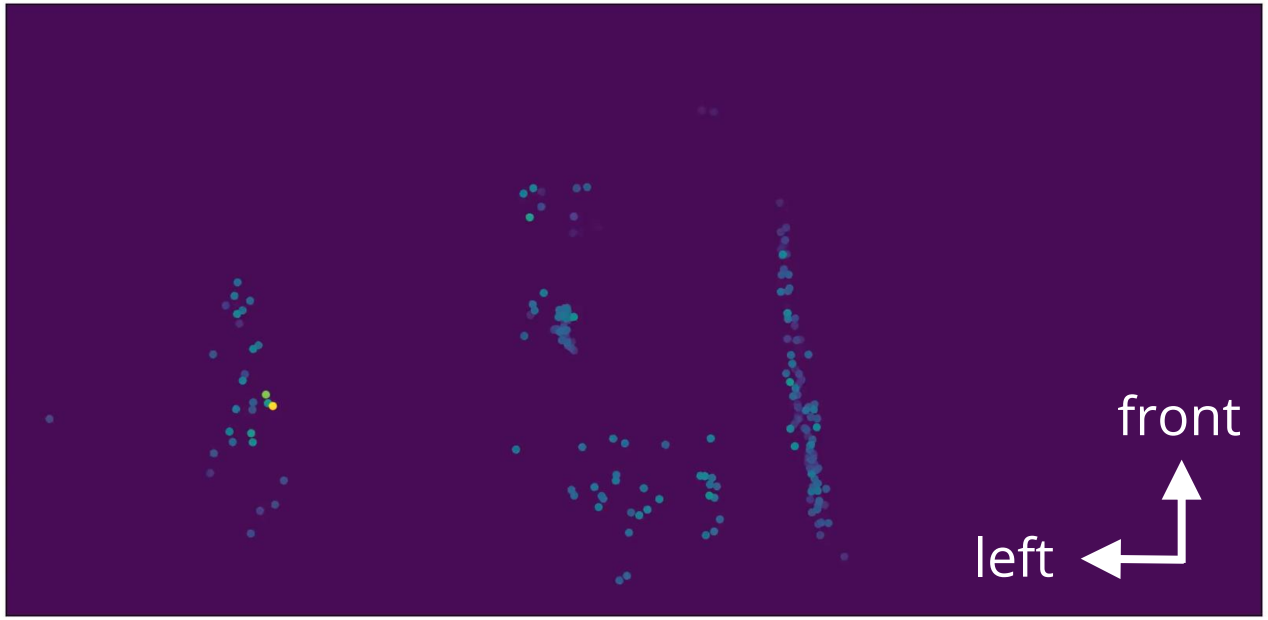
Radar Processing: **Angle of Arrival Estimation**



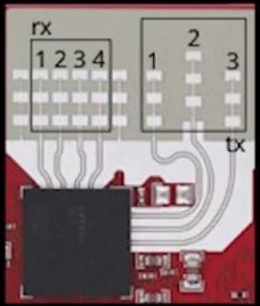


Radar Point Clouds
Poor Resolution & Low Density

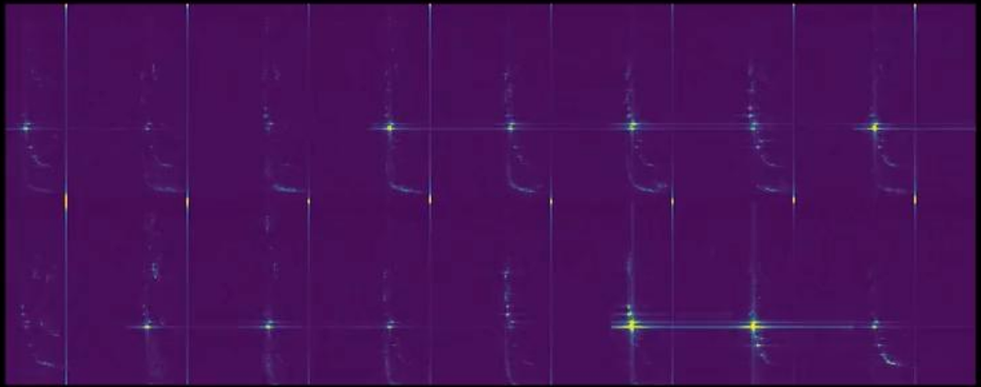




Input: 4D Radar Spectrum

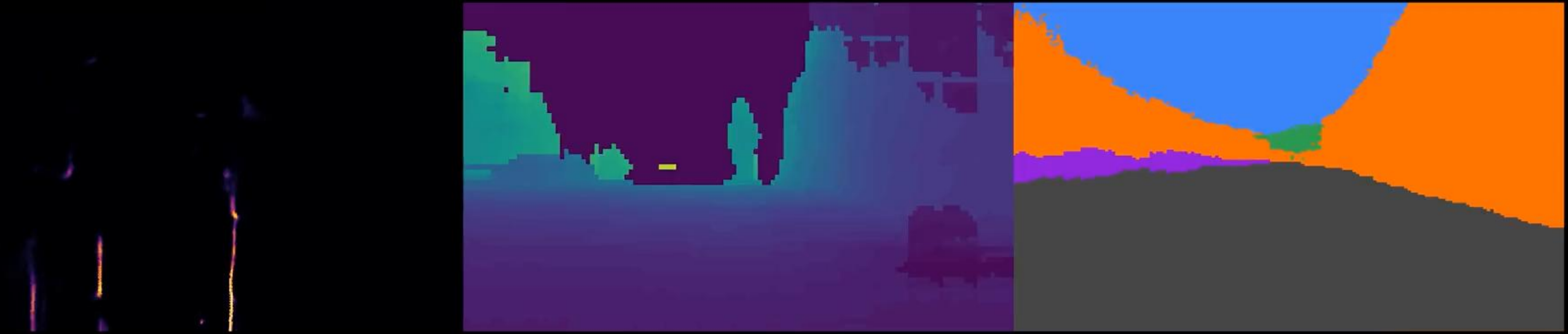


Chip Radar
3x4 Antenna

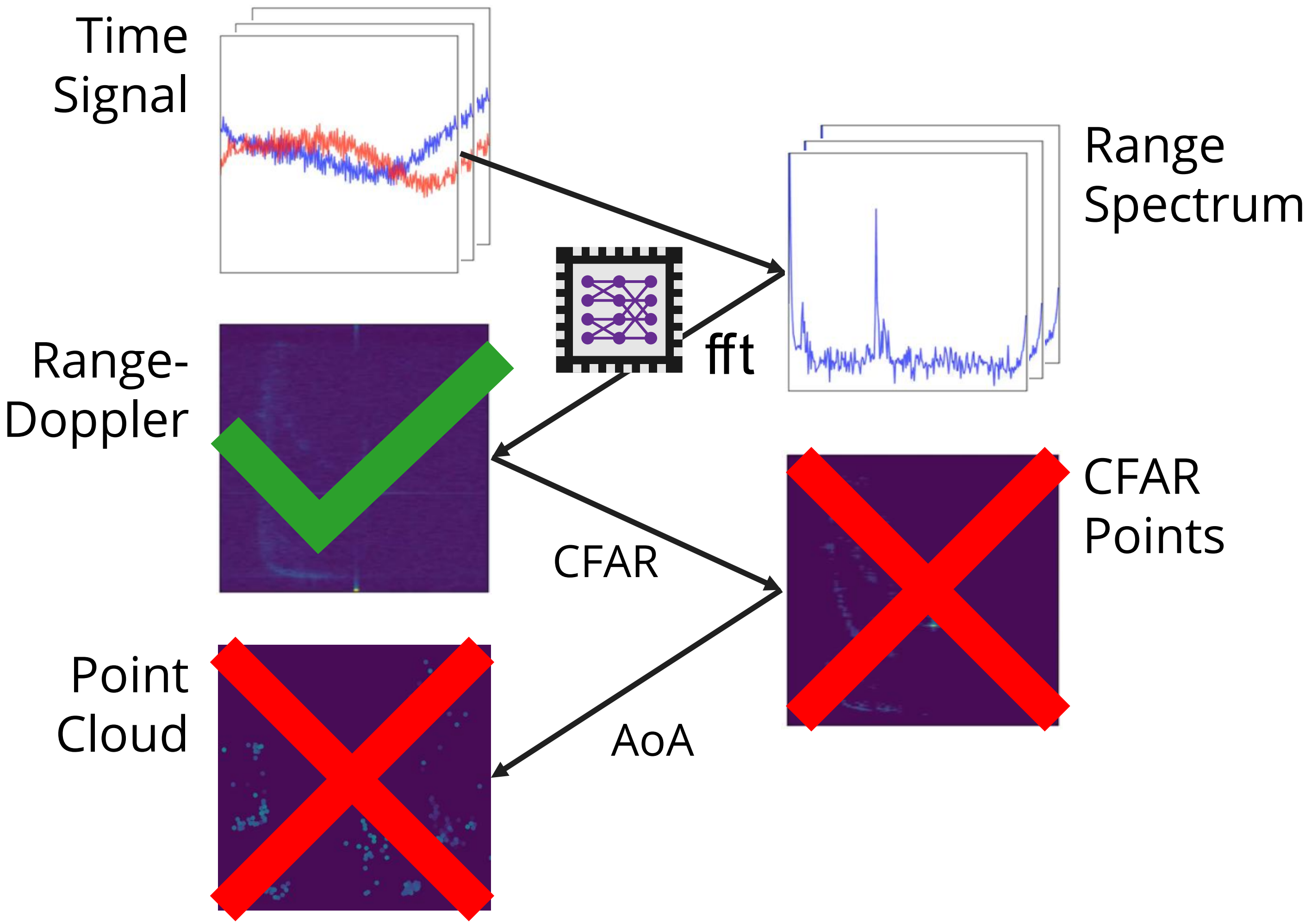


Reference Scene

Our GRT Output (Using Only 4D Radar Input)

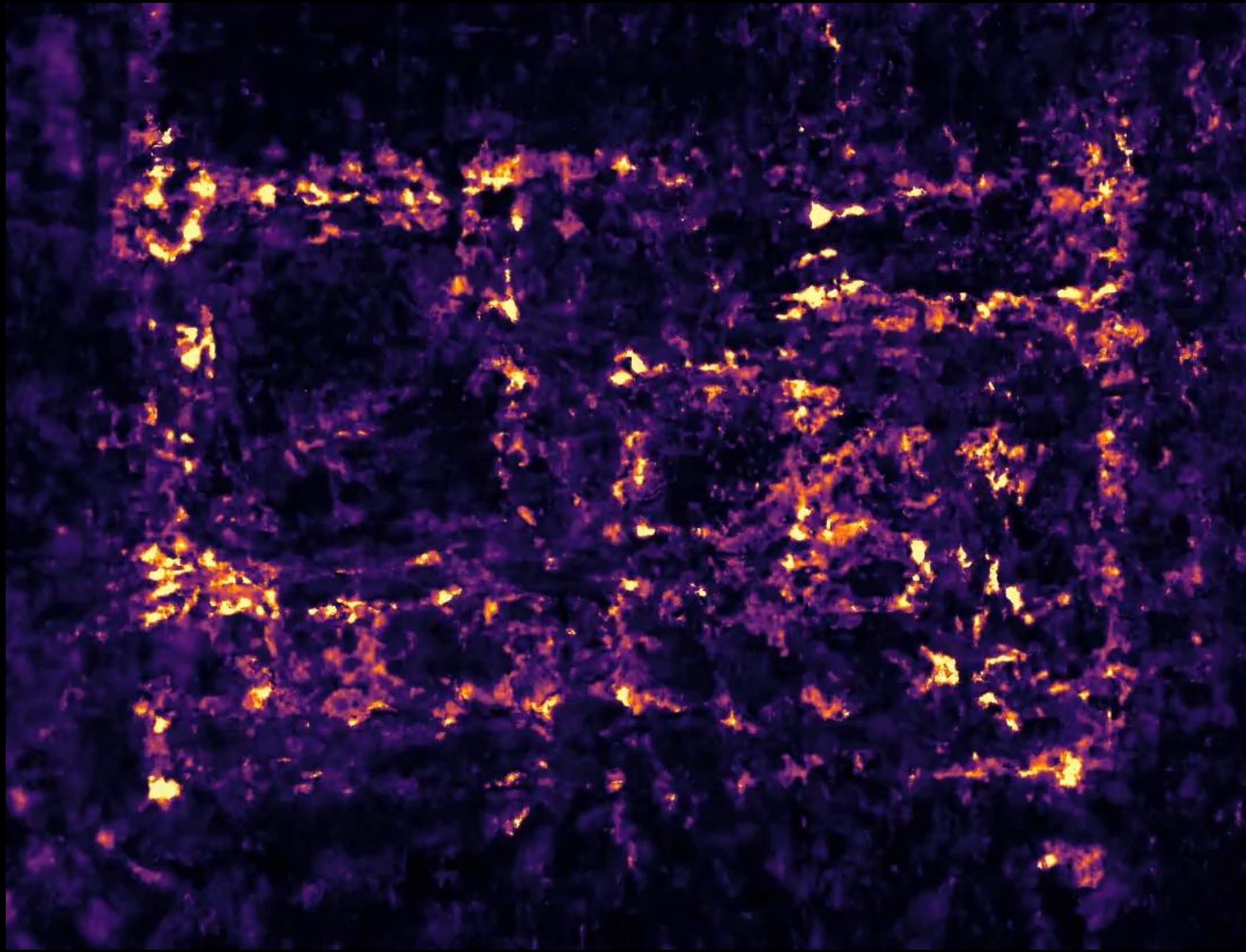


Key Insight #1: **use “raw” unfiltered spectrum**



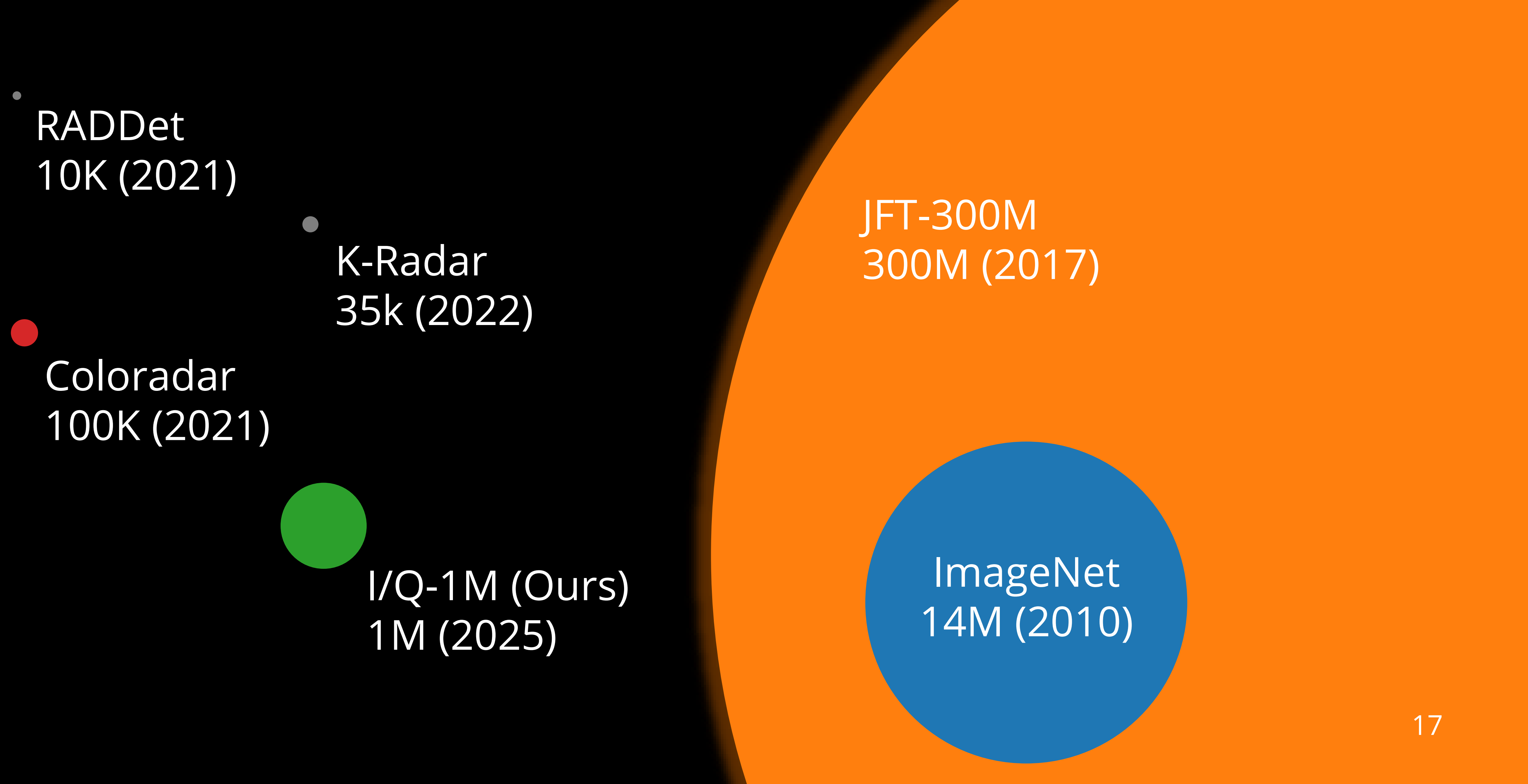
Key Insight #1: **use “raw” unfiltered spectrum**

CFAR – Point Cloud Aggregation



DART (CVPR ‘24) – Dense Tomography via Range-Doppler Spectrum

Key Insight #2: **collect more data**



So we started scaling...

I/Q-1M Dataset



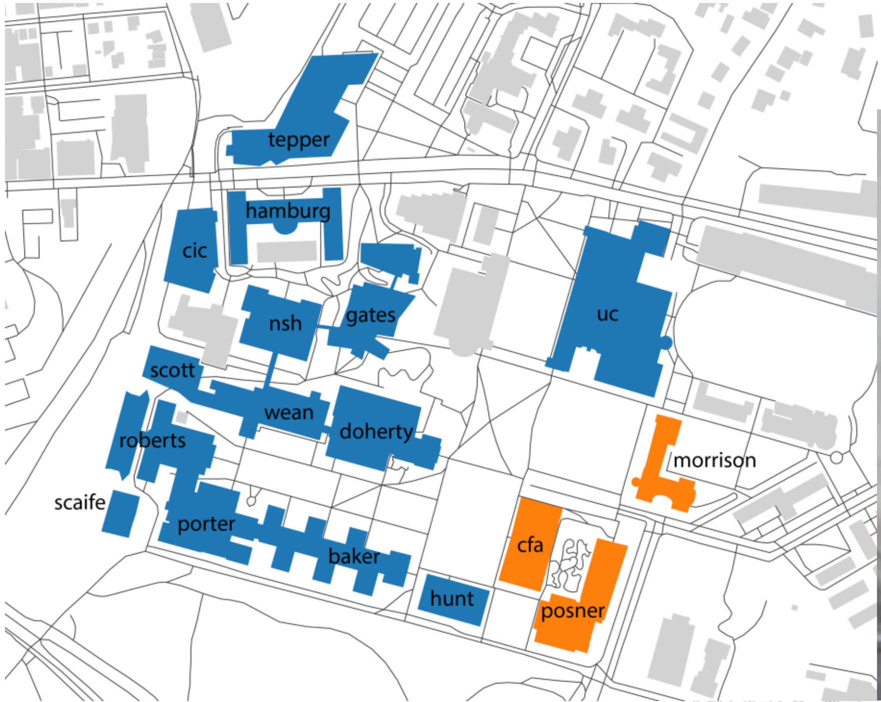

I/Q-1M Dataset

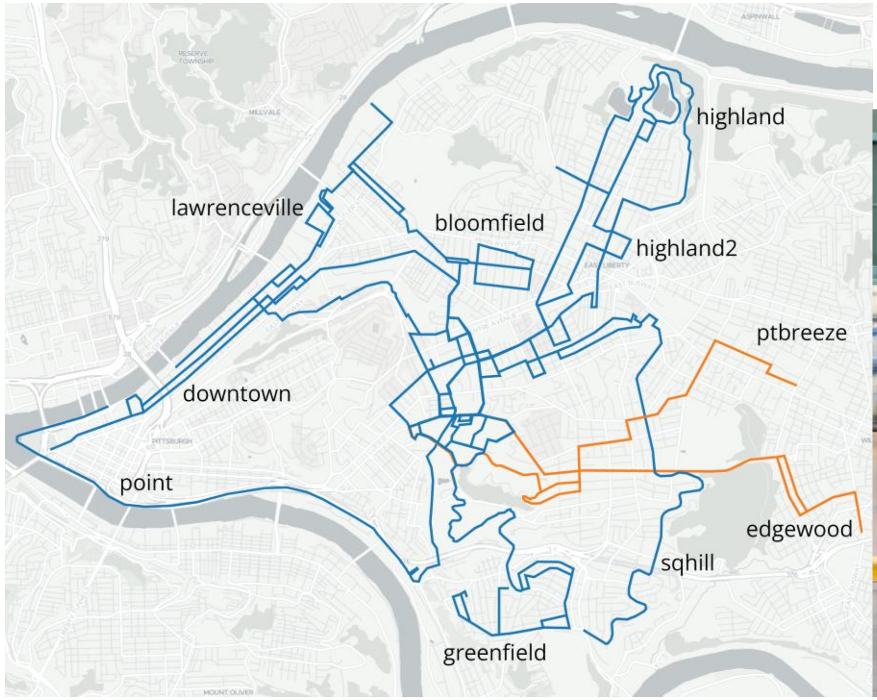
I/Q-1M Dataset

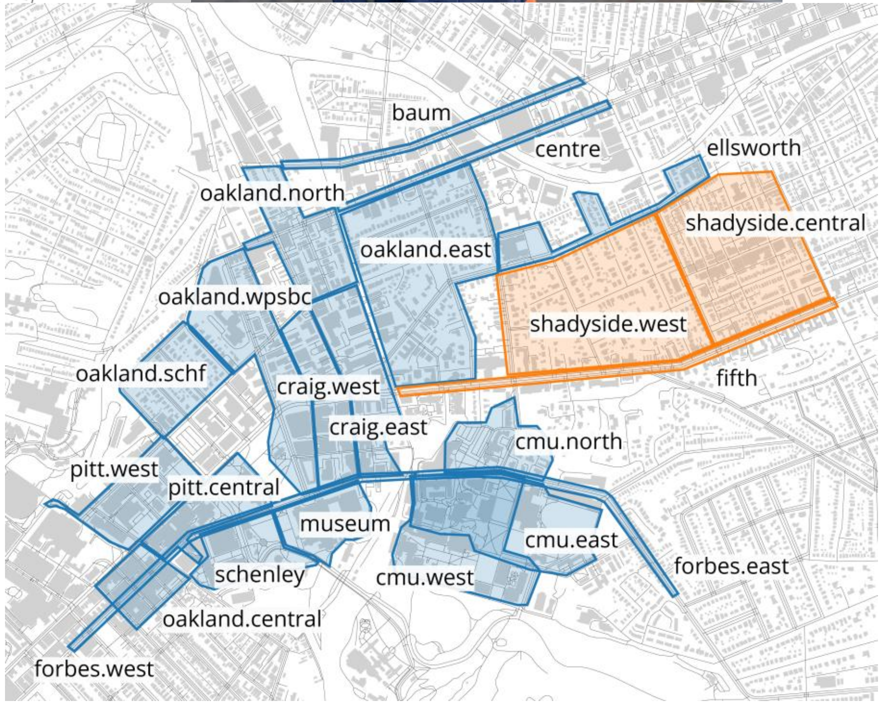
I/Q-1M Dataset

I/Q-1M Dataset

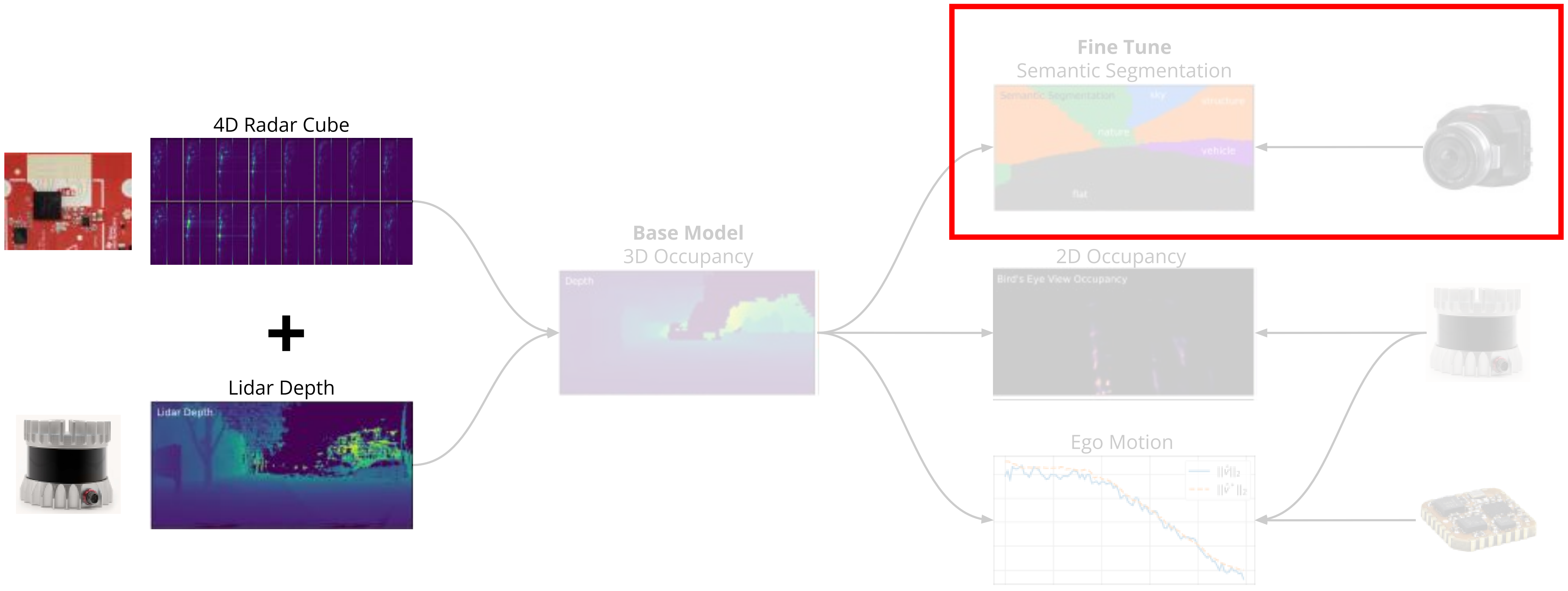
- 29 hours of data
- 1M radar-lidar-camera samples

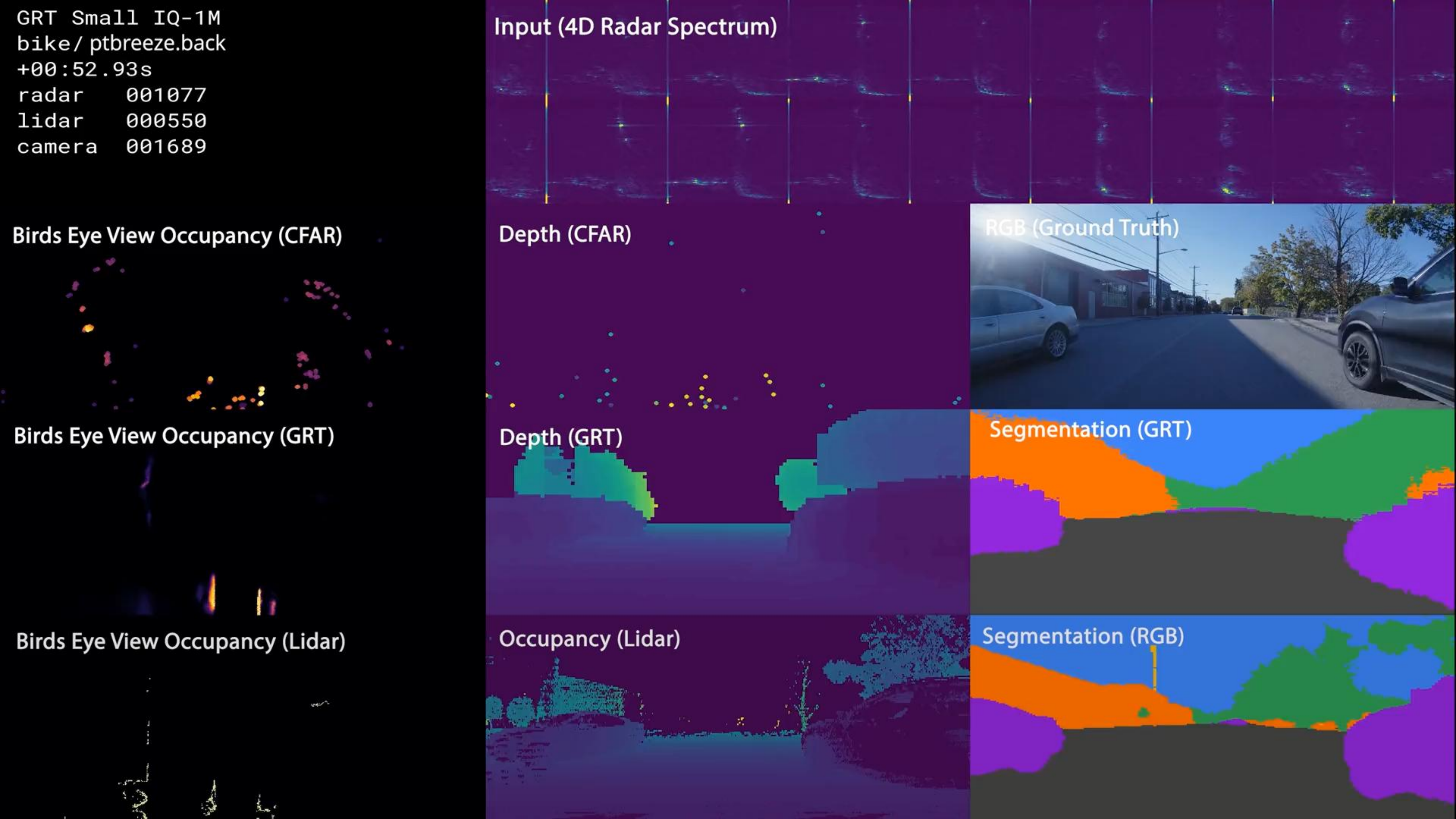


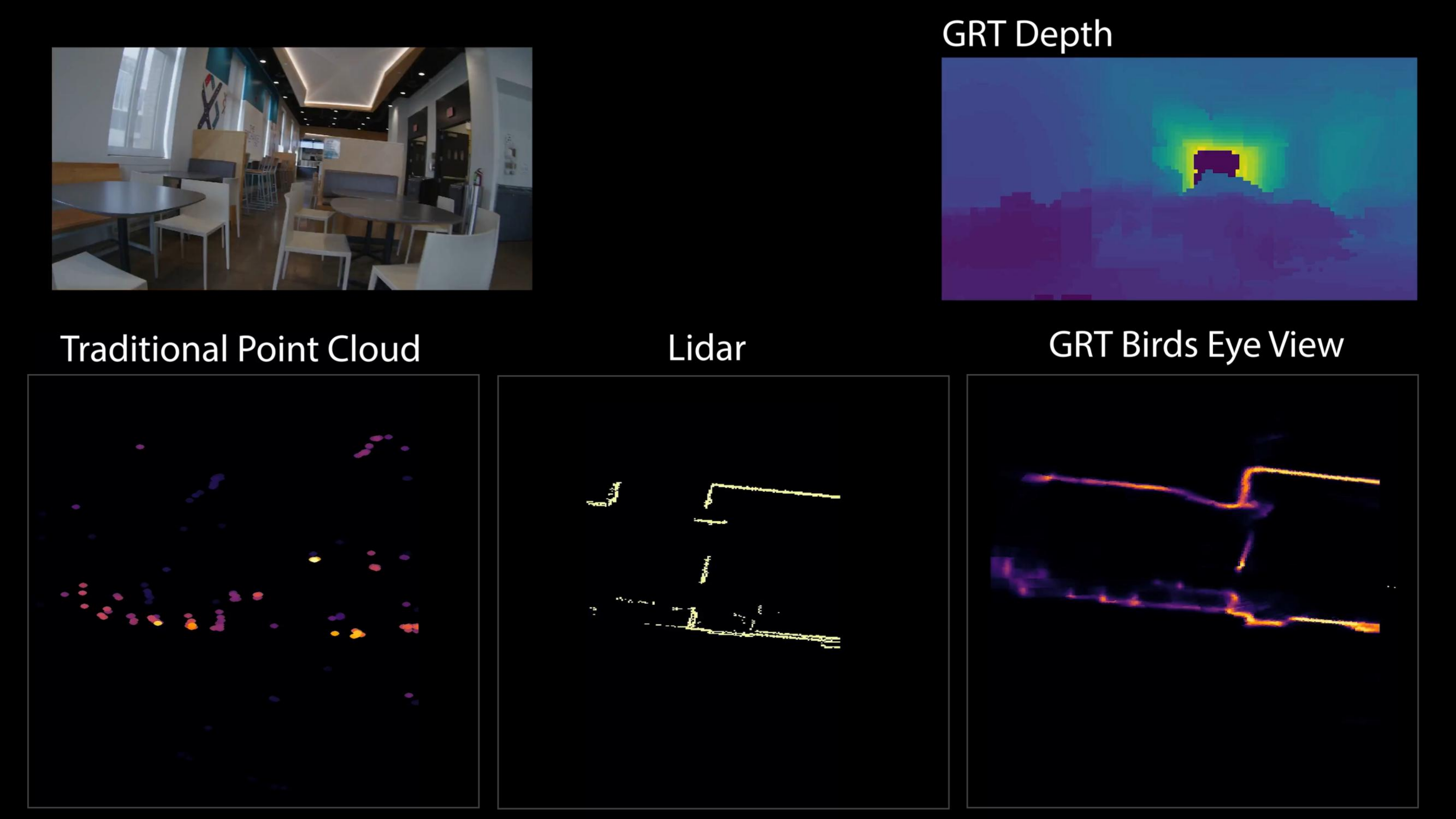




Generalizable Radar Transformer

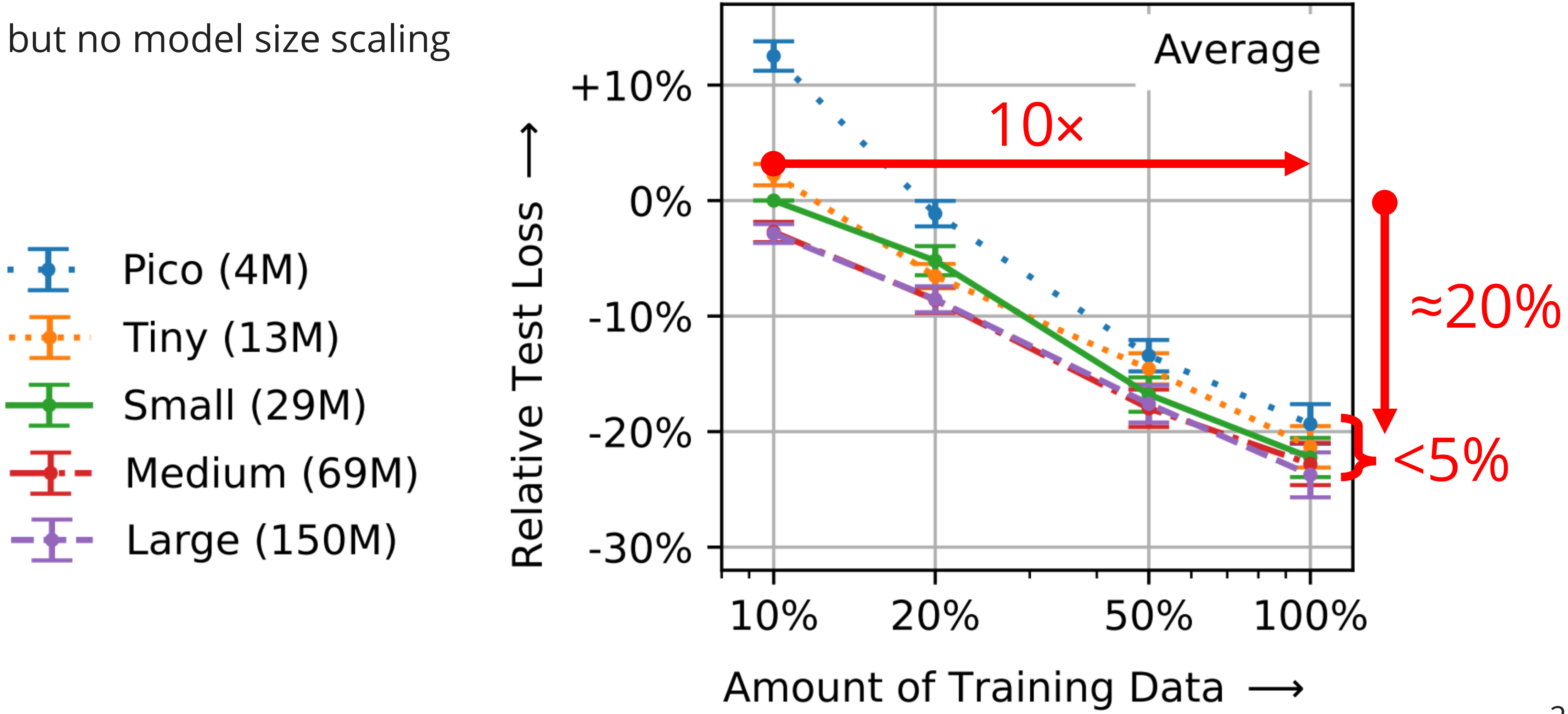



GRT Small IQ-1M
bike/ ptbreeze.back
+00:52.93s
radar 001077
lidar 000550
camera 001689
Input (4D Radar Spectrum)
Birds Eye View Occupancy (CFAR)
Depth (CFAR)
RGB (Ground Truth)
Birds Eye View Occupancy (GRT)
Depth (GRT)
Segmentation (GRT)
Birds Eye View Occupancy (Lidar)
Occupancy (Lidar)
Segmentation (RGB)

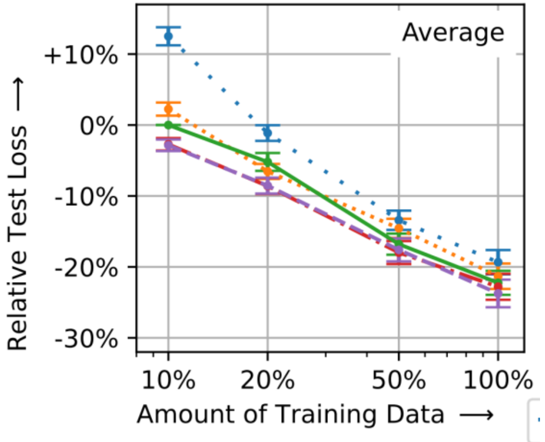

GRT Depth
Traditional Point Cloud
Lidar
GRT Birds Eye View

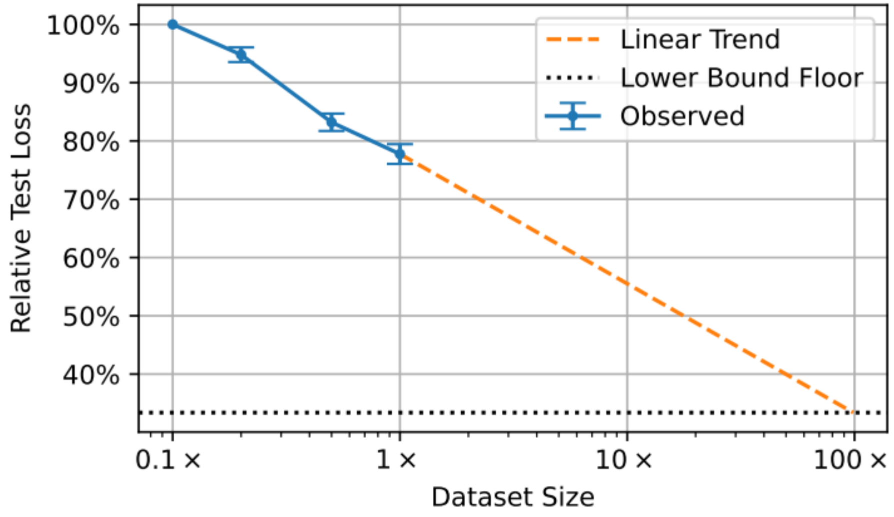
Key Results: **Logarithmic Scaling**

… but no model size scaling



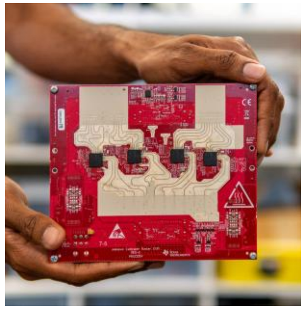
Key Results: ~~Towards~~ **Foundational Models?**

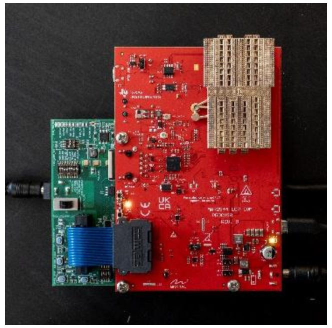




What Next? **Keep Scaling!**

- More sensors
- More settings
- More data





Please reach out if interested!

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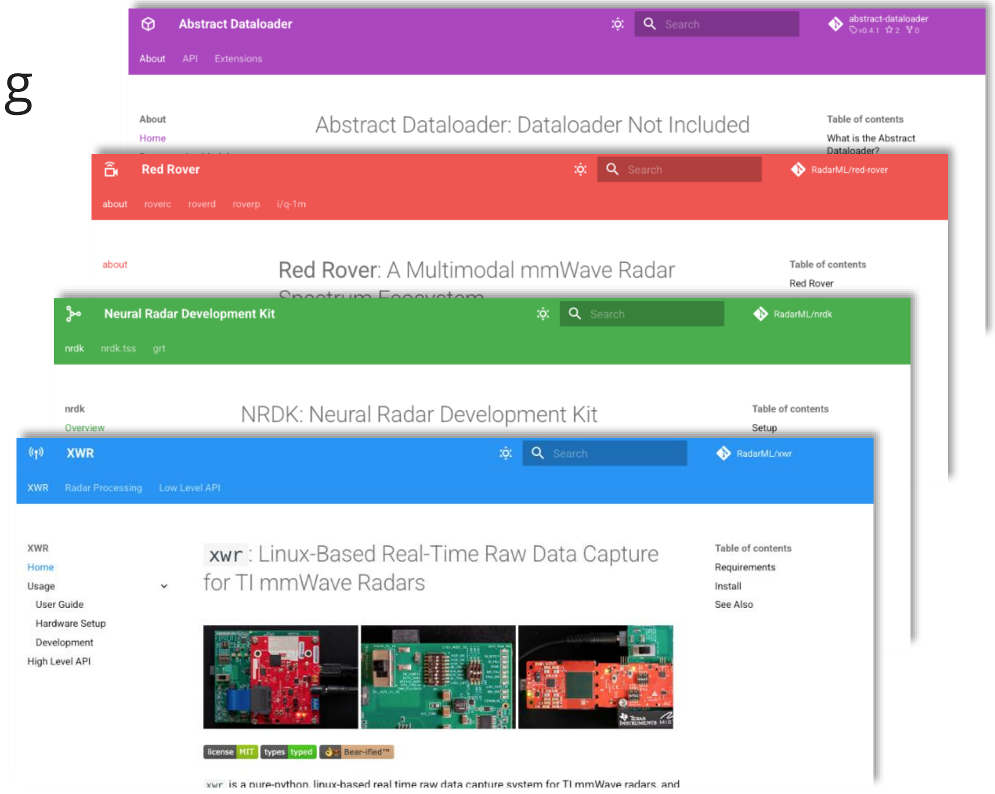
What Next? The **RadarML Ecosystem**

- Full research ecosystem for learning on radar spectrum
- Tools, datasets, libraries …
- Permissively licensed



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Towards Foundational Models for Single-Chip Radar

Project Site

wiselabcmu.github.io/grt

RadarML

radarml.github.io